

Andy Yang

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EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science in Electrical Engineering, Minor in Semiconductor Engineering

Expected: December 2028

GPA: 3.96/4.00

- Honors: Dean's List, James Scholar

Relevant Course Work: Computer Systems & Programming, Analog Signal Processing, Digital Systems Laboratory, Photonic Devices

WORK EXPERIENCE

Bionanophotonics Research Lab

Undergraduate Researcher

Urbana-Champaign, IL

January 2026 – Present

- Designed and fabricated custom PCBs for Tumor Treating Fields applications, transitioning amplifier architecture to higher-bandwidth op-amps with an added voltage follower stage to ensure signal stability.
- Identified, diagnosed, and resolved layout/circuit errors across custom board revisions through schematic review, board fabrication, and hardware validation.
- Enhanced circuit stability and reduced output signal noise by integrating decoupling capacitors, optimizing feedback resistor values, and testing modifications on physical breadboards/prototypes.

Illini Solar Car

Electrical Sub-team Member & Media Lead

Urbana-Champaign, IL

August 2025 – Present

- Led development of WaveSculptor motor control and power conversion system, enabling low-voltage DC solar power to efficiently drive AC M1096-III motors and sensors through optimized inverter design and thermal management.
- Engineered high-efficiency solar arrays by organizing Gocherman solar cell layouts and strategically grouping modules with Maximum Power Point Trackers to optimize power generation.
- Fabricated battery packs via spot welding for Project Delos, optimizing cell performance and safety through rigorous cell characterization, precise state-of-charge balancing, and capacity binning.
- Immersed in multi-day technical scrutineering operations, synthesizing regulations, safety compliance, and engineering designs across 20+ collegiate solar car teams

IDEAL Summer Program at California State University Fullerton

Mechanical Engineering Research Intern

Fullerton, CA

June 2024 – August 2024

- Developed kinematic models in GeoGebra to analyze and plot Theo Jansen linkage motion, optimizing foot trajectories that mimic human gait and enable rover mobility over terrain unsuitable for tracked or wheeled systems.
- Applied model-derived data to design and construct a legged rover prototype incorporating Theo Jansen linkages, validating simulated motion through physical implementations.
- Presented interdisciplinary research at the 2025 California State University Biotechnology Symposium, integrating biomechanics, robotic system design, and computational modeling inspired by biological locomotion.

World Relief Chamber Music

Co-President & Head of Peer-to-Peer Fundraising

Chino Hills, CA

August 2022 – January 2025

- Managed a peer-to-peer donation system that raised over \$25,000 in humanitarian donations for the Russo-Ukrainian War, the Sudanese Civil War, and the 2023 Hawaii wildfires.
- Coordinated weekly orchestral and classical ensemble rehearsals for charitable holiday performances at senior living centers.

PROJECTS

Watt's Up Sun?

- Engineered an analog circuit schematic for a device that follows the sun without the need for modern digital components, utilizing op-amps and other analog equipment to create subtractors, comparators, and voltage followers.

Boe-Bot Maze Navigator

- Programmed infrared signals using the P-Basic language and manufactured physical chassis using SolidWorks to create an intelligent robotic cargo carrier capable of navigating through mazes.

Vending Machine Controller

- Designed and implemented a vending machine controller in Vivado, using logic gates and flip-flops for state control and data retention
- Developed testbenches to simulate, debug, and verify correct operation across input scenarios
- Deployed the design on a physical protoboard, validating real-world functionality and hardware integration

Applause Meter

- Designed an analog sound-detection circuit using a microphone and amplifier to reliably detect clap events.
- Used comparator thresholds to distinguish differing clap events to activate corresponding LED outputs.
- Iteratively devised multiple solutions and analyzed relative design tradeoffs to achieve the best possible performance.

TECHNICAL SKILLS

Programming Languages: Python, P-Basic, TI-Basic, C/C++, LC3

Applications: SolidWorks, LTspice, GitHub, Prohelion Profinity, Vivado, KiCad, Word, Excel, Powerpoint